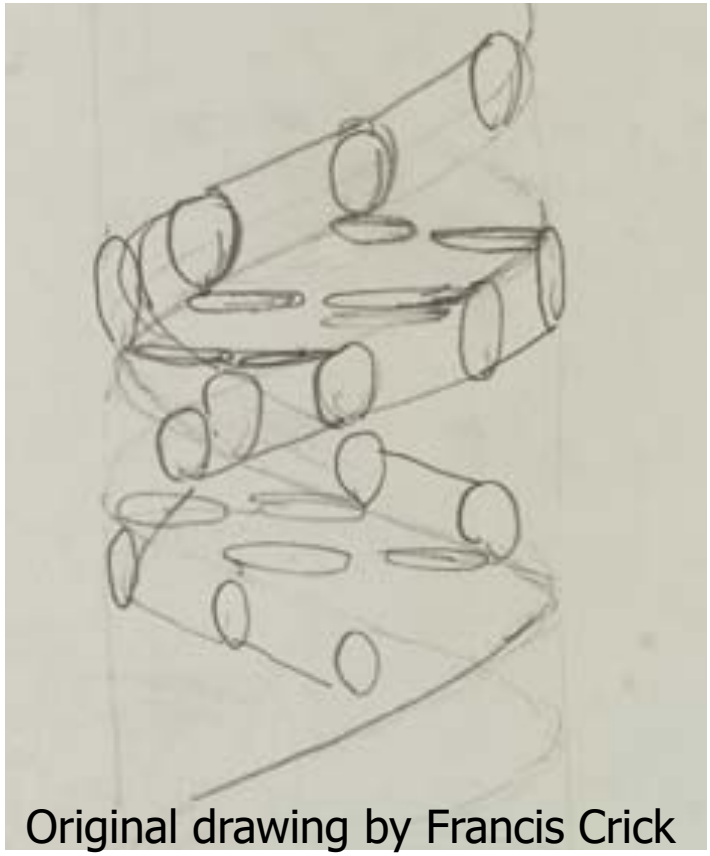
Lecture 7.3.

DNA replication

Dr. Matias Avila
maavila@unav.es

DNA REPLICATION

"It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material"



Original drawing by Francis Crick

Watson & Crick

Nature (1953)

DNA replication

Scientists wondered at the ability of

1. Organisms to create **copies** of themselves
2. Cells to produce **identica**l copies of large macromolecules.

Speculation centered around the concept of a **template:**

1. **Structure** that would allow molecules to be lined up to create a macromolecule with a unique sequence and function.
2. **The** 1940s brought the revelation that **DNA** was the genetic molecule.

James Watson and Francis Crick deduced the way in which DNA could act as a template:

1. One strand is the **complemen**t of the other.
2. Each strand provides the template for a sister strand with a predictable and **complementary** sequence.

DNA Replication Follows a Set of Fundamental Rules.

Early research on bacterial DNA replication and its enzymes established properties that have proven applicable to DNA synthesis in every organism.

The fundamental properties of the DNA replication process have proved to be essentially identical in all species.

1. ***DNA replication is semiconservative.***

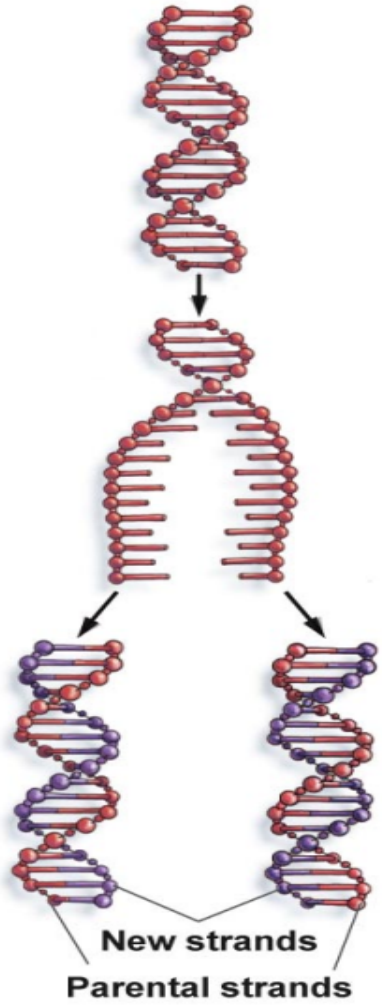
2. ***Replication begins at an origin and usually proceeds bidirectionally.***

3. ***DNA synthesis progresses in the 5' to 3' direction and is a semi-discontinuous process.***

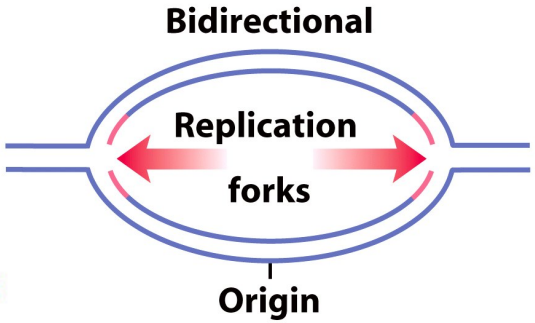
1. DNA Replication Is Semiconservative.

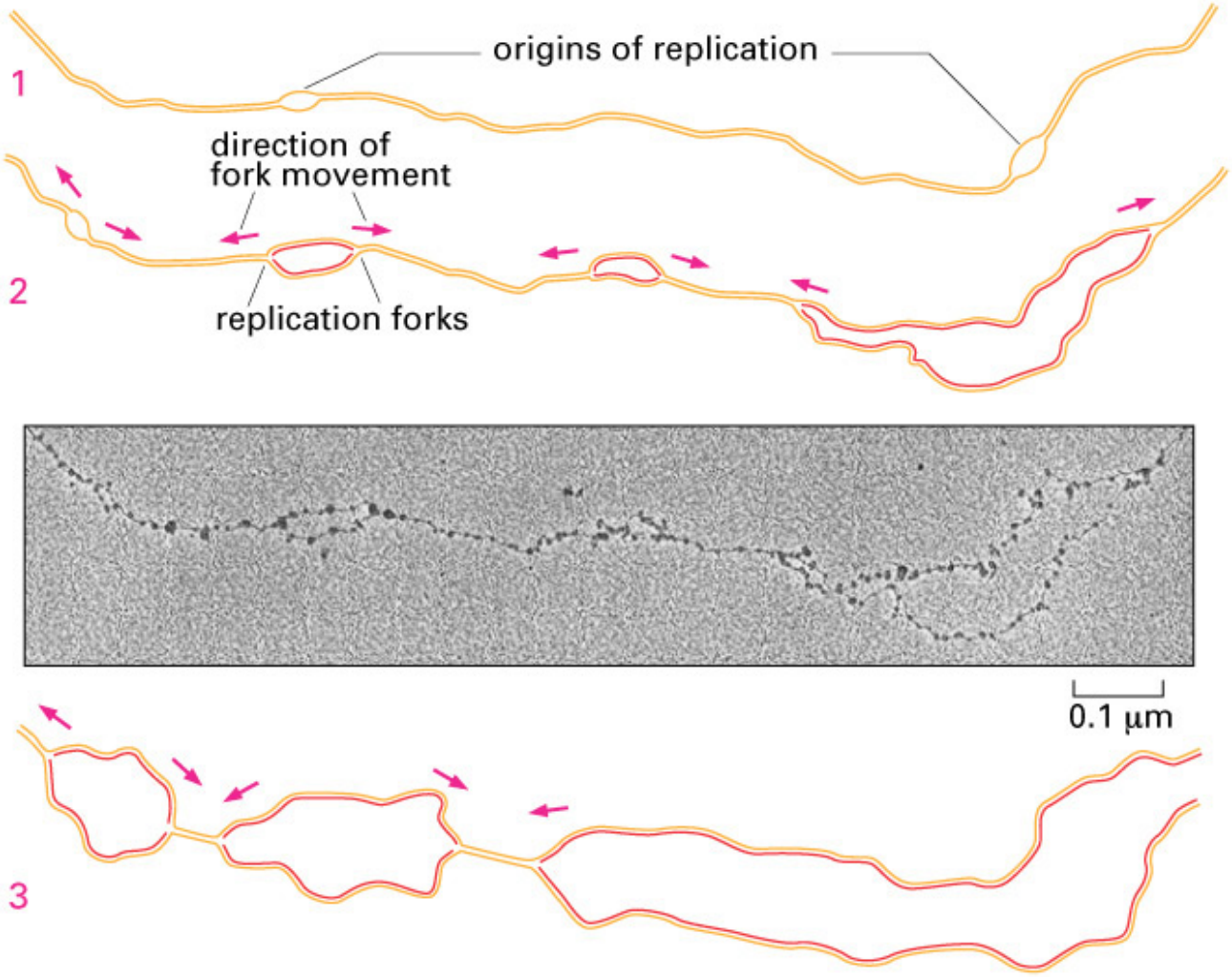
Each DNA strand serves as a template for the synthesis of a new strand, producing two new DNA molecules, each with one new strand and one old strand.

Watson and Crick proposed the hypothesis of semiconservative DNA replication.
The hypothesis was proved by
Matthew Meselson and Franklin Stahl in 1957.

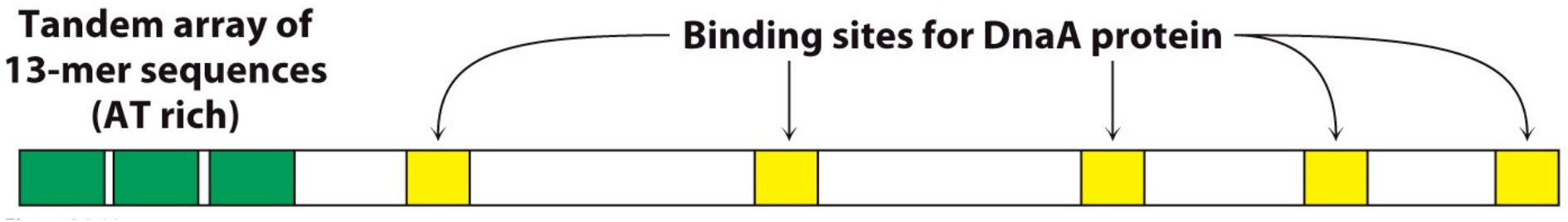


2. *Replication begins at an origin and proceeds bidirectionally.*





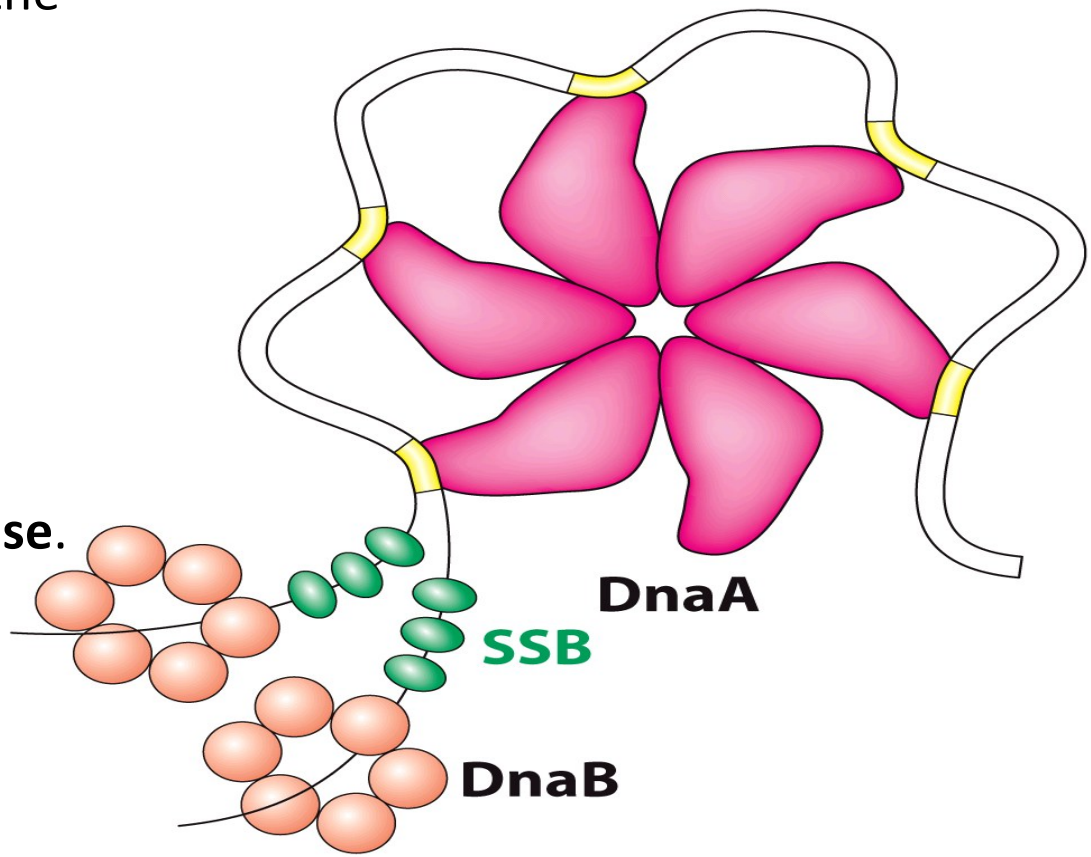
Origin of replication in *E. coli*: a 245 bp region termed oriC



DnaA hexamer binding to oriC allows the recruitment of the **helicase DnaB**.

Single stranded DNA is then bound by the **single stranded DNA binding proteins (SSB)**.

This complex facilitates the binding of **primase** (**DnaG**) and **DNA polymerase**.



Origin of replication in eukaryotes

Although the overall mechanism of DNA replication in eukaryotes and bacteria is conserved. it is significantly complex in nucleated cells due to:

1. The size of the genome ($4x10^6$ *vs* $6x10^9$ bps).
2. The number of chromosomes (1 *vs* 23 pairs in humans).
3. The *E. coli* chromosome is a circular molecule, while human chromosomes are linear.

In eukaryotes there are **multiple origins of replication** ($3x10^5$ in humans).

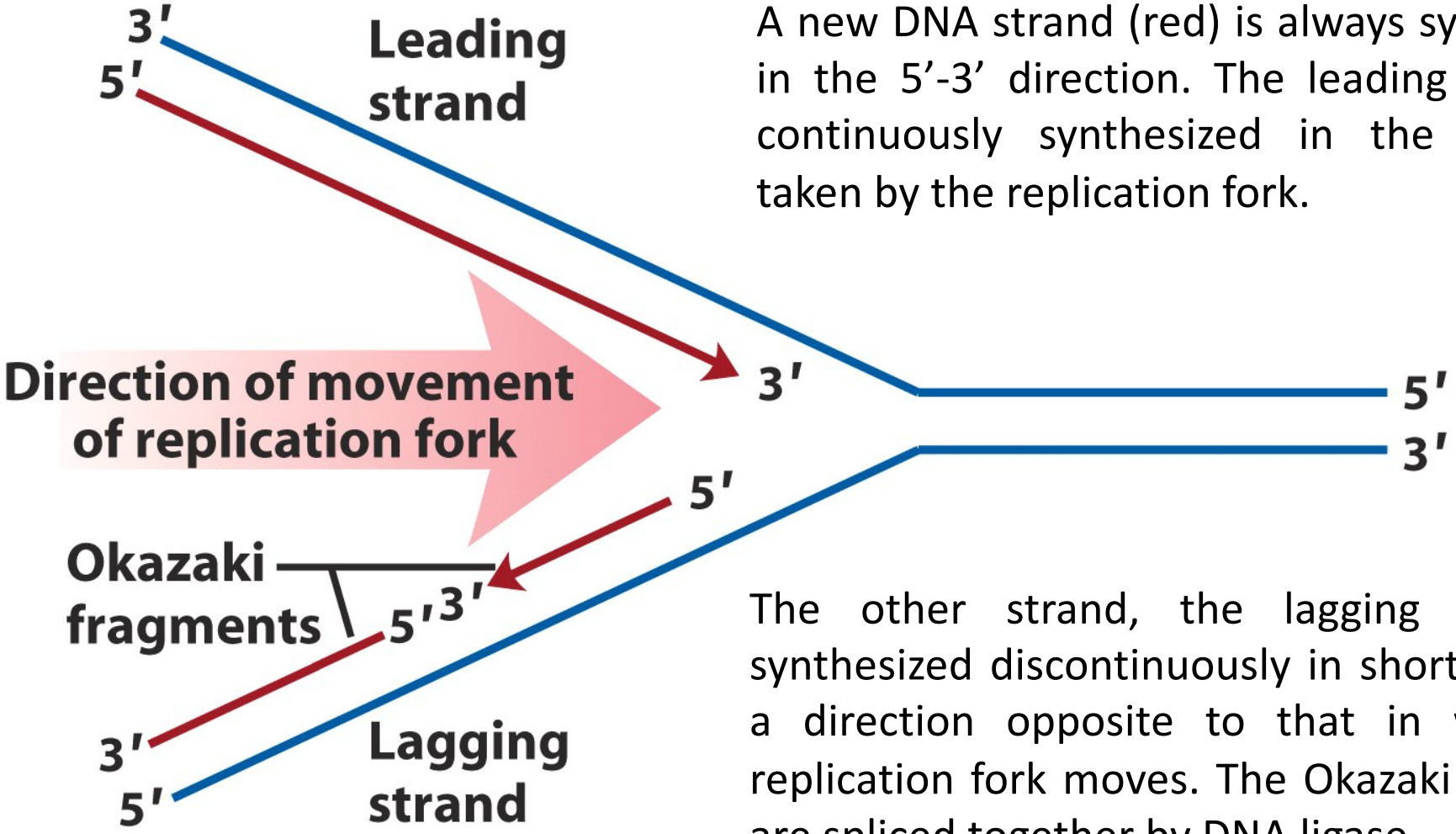
There are **no specific DNA sequences** associated with these origins of replication.

Origins of replication usually are AT-rich regions around which the **origin of replication complexes** (**ORC**) associate.

ORCs include, among other auxiliary proteins:

- An **hexamer complex** similar to that formed by bacterial **DnaA**.
- A **helicase complex** formed by six subunits of the **Mcm2-7** protein.
- A single-stranded DNA binding protein known as "**replication protein A**".
- Two different **polymerases**, **DNA pol** α and **DNA pol** δ, that sequentially interact with the ORC.

3. DNA synthesis proceeds in the 5’ to 3’ direction and is a semi-discontinuous process.

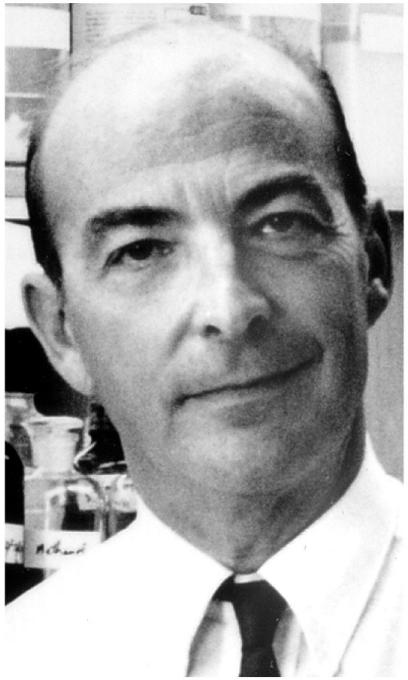


A new DNA strand (red) is always synthesized in the 5’-3’ direction. The leading strand is continuously synthesized in the direction taken by the replication fork.

The other strand, the lagging strand, is synthesized discontinuously in short pieces, in a direction opposite to that in which the replication fork moves. The Okazaki fragments are spliced together by DNA ligase.

In bacteria Okazaki fragments are ≈ 1000 to 2000 nucleotides long.
In eukaryotes they are 150-200 nucleotides long.

DNA POLYMERASES



Arthur Kornberg

The search for an enzyme that could synthesize DNA began in 1955.

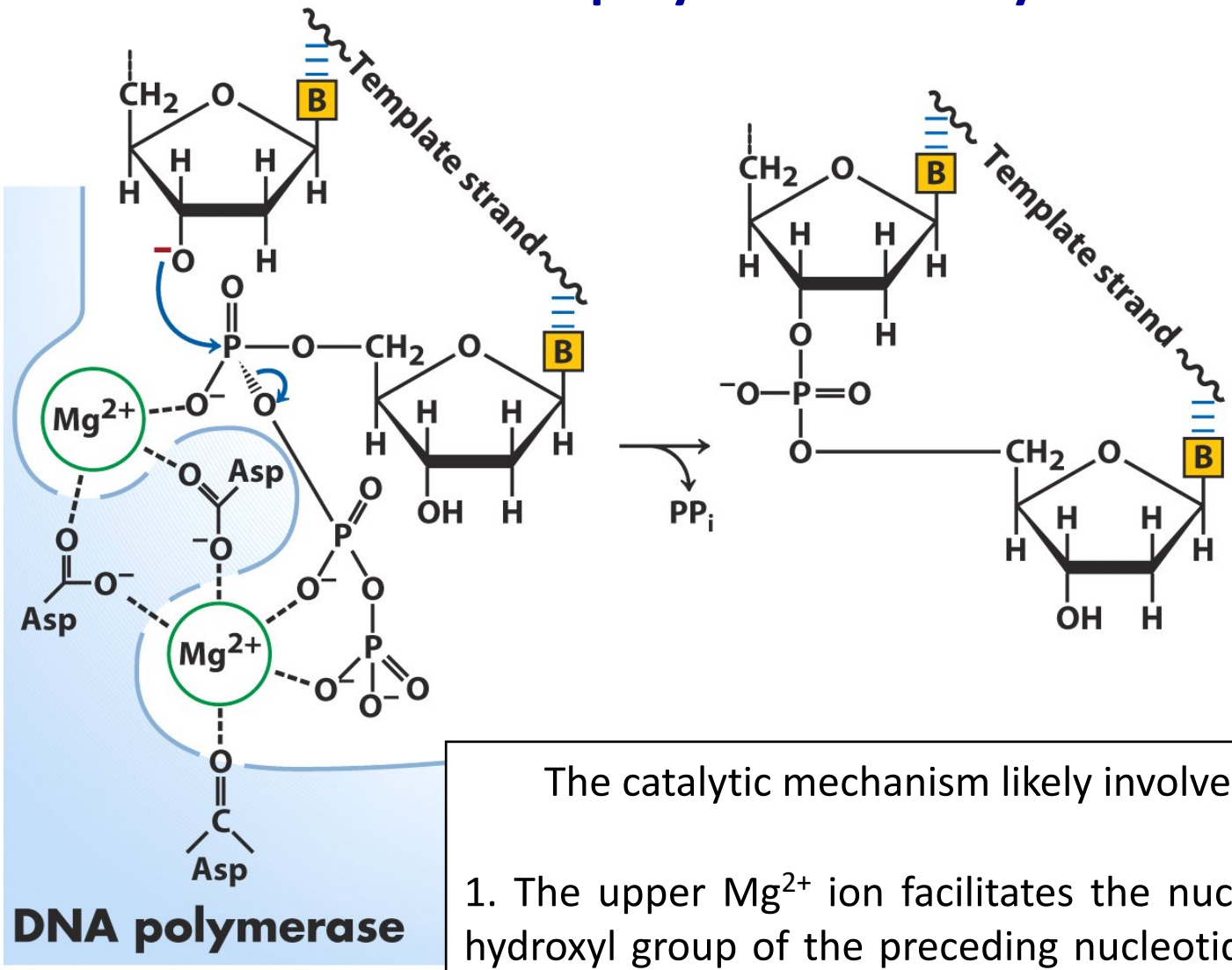
Work by Arthur Kornberg led to the purification and characterization of DNA polymerase from *E. coli*, a single-polypeptide enzyme now called **DNA polymerase I**.

Much later, investigators found that *E. coli* contains **at least four other distinct DNA polymerases**.
The essential mechanism of the DNA synthetic reaction is conserved among species.

There are two basic requirements for DNA polymerization:

1. All DNA polymerases require a **template**. Watson and Crick's base pairing dictates the sequence of the new DNA strand.
2. A **primer** with a free 3'-OH end is needed to start the reaction. These primers are **RNA oligonucleotides** (of about 5 nucleotides) synthetized by and RNA polymerase (primase).

DNA polymerase catalytic mechanism

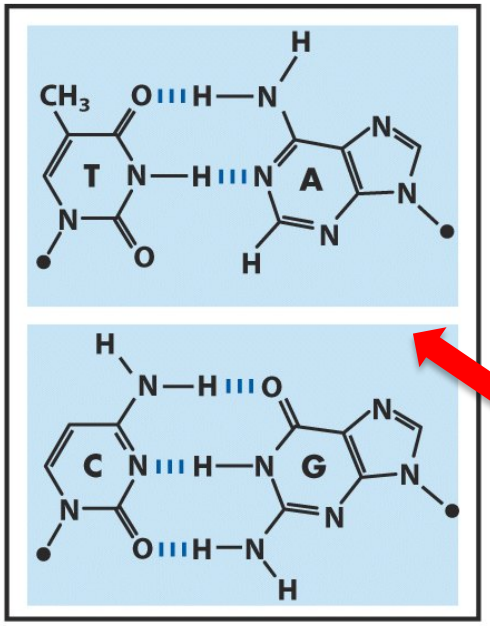


The catalytic mechanism likely involves two Mg^{2+} ions.

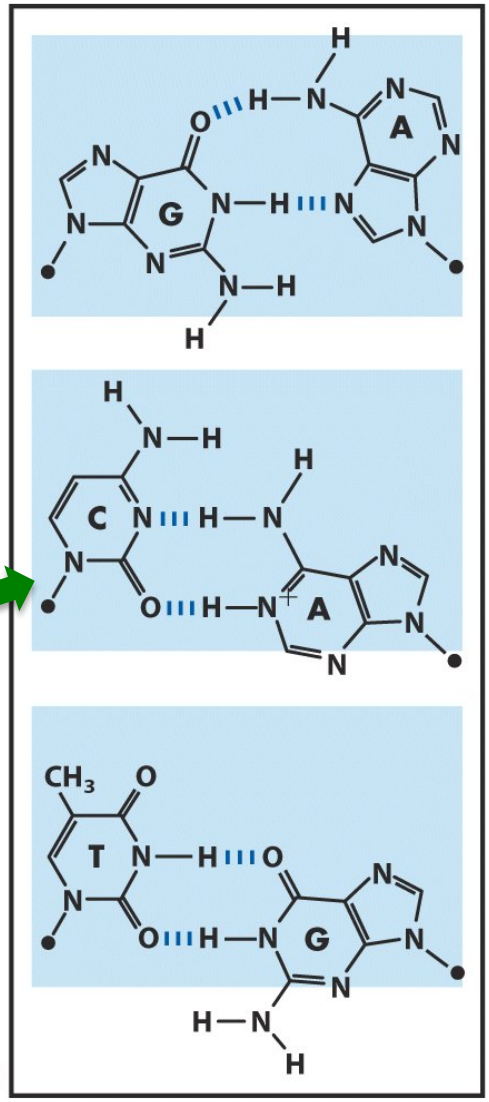
1. The upper Mg^{2+} ion facilitates the nucleophyllic attack of the 3-hydroxyl group of the preceding nucleotide on the α–phosphate of the incoming nucleotide triphosphate.
2. The lower Mg^{2+} ion facilitates displacement of the pyrophosphate.

Both ions stabilize the structure of the transition state. RNA polymerases use a similar mechanism.

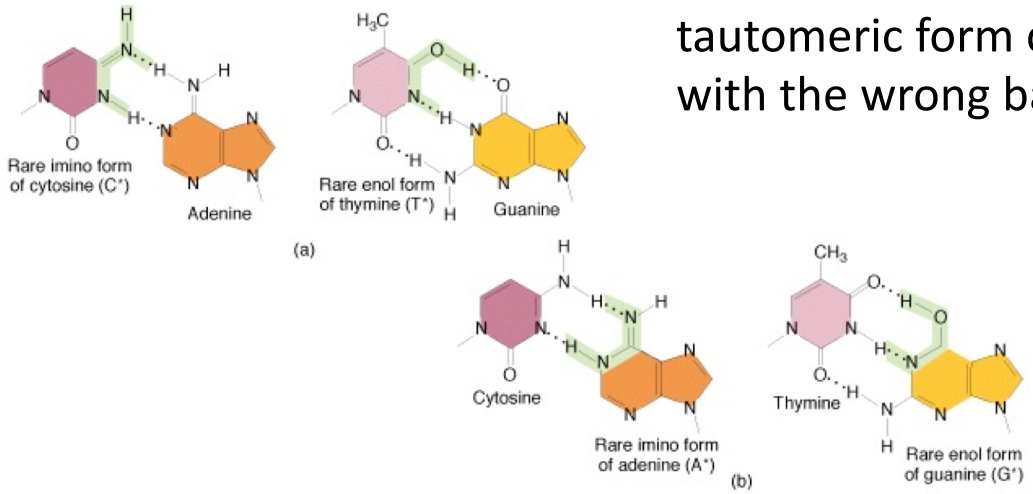
DNA REPLICATION IS A VERY ACCURATE PROCESS

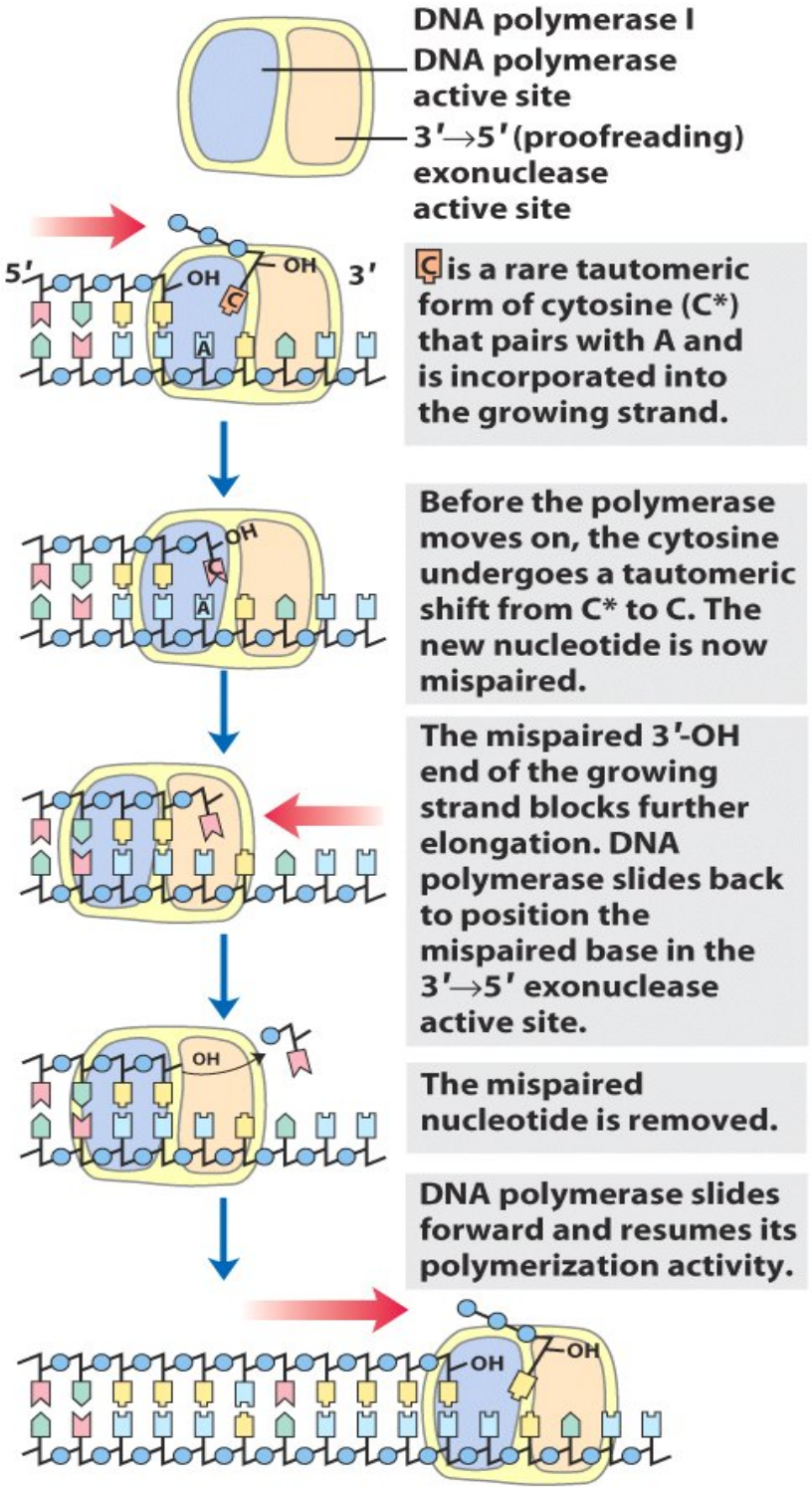


- Nucleotide selection is not only based on the formation of hydrogen bonds, it is also influenced by the **geometry of the base pairs**.
- The standard A-T and G-C base pairs have similar geometries, and an active site will accommodate both.
- The geometry of incorrectly paired bases can exclude them from the active site.



Selection o bases in the incorrect tautomeric form can lead to pairing with the wrong base.





DNA REPLICATION IS A VERY ACCURATE PROCESS

1. *In vitro* studies showed that DNA polymerases insert one incorrect nucleotide for every 10^4 to 10^5 correct ones.

2. These mistakes occur because a base is briefly in an unusual tautomeric form, allowing it to hydrogen-bond with an incorrect partner.

3. In vivo, the error is reduced by enzymatic mechanisms.

4. If a polymerase has added the wrong nucleotide, translocation of the enzyme to the position where the next nucleotide is to be added is inhibited.

5. The 3-5 exonuclease activity removes the mispaired nucleotide, and the polymerase begins again.

This activity is known as **proofreading**, but it is not the reverse of the polymerization reaction.

Base selection + proofreading activity reduce the errors to one in 10^6 to 10^8 added base pairs. Additional repair systems further reduce this rate.

DNA polymerases in *E. coli*

	DNA polymerase		
	I	II	III
Structural gene*	*polA*	*polB*	*polC* (*dnaE*)
Subunits (number of different types)	1	7	≥10
M_r	103,000	88,000†	791,500
3′→5′ Exonuclease (proofreading)	Yes	Yes	Yes
5′→3′ Exonuclease	Yes	No	No
Polymerization rate (nucleotides/s)	16–20	40	250–1,000
Processivity (nucleotides added before polymerase dissociates)	3–200	1,500	≥500,000

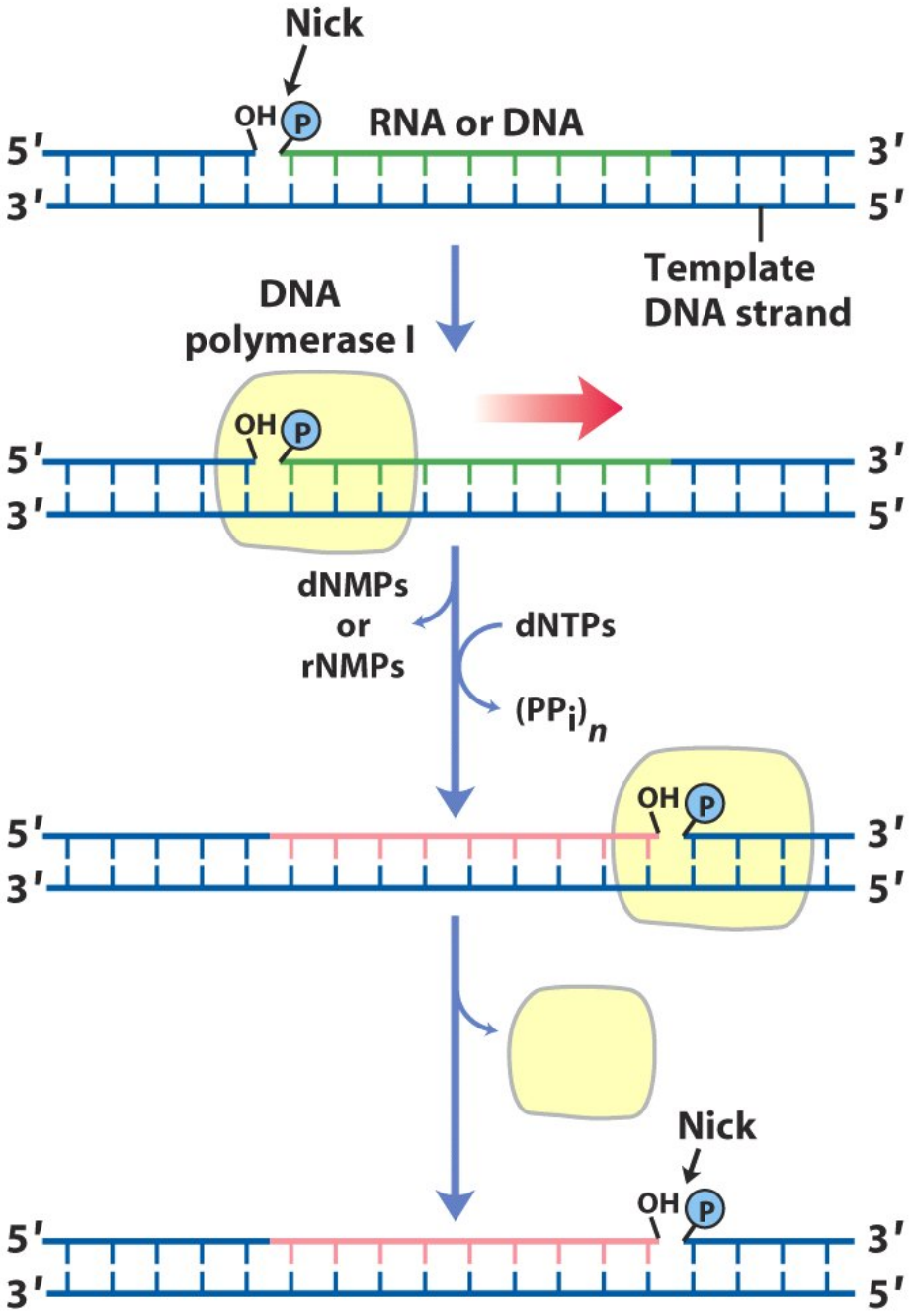
*For enzymes with more than one subunit, the gene listed here encodes the subunit with polymerization activity. Note that *dnaE* is an earlier designation for the gene now referred to as *polC*.

†Polymerization subunit only. DNA polymerase II shares several subunits with DNA polymerase III, including the β, γ, δ, δ′, χ, and ψ subunits (see Table 25–2).

1. More than 90% of the DNA polymerase activity observed in *E. coli* can be accounted for by DNA polymerase I.
2. Other DNA polymerases in *E. coli* include DNA polymerase II and III.
3. DNA Polymerase I is mainly involved in repair, recombination and “clean up” functions during replication (functions enhanced by its 5’-3’ exonuclease activity).
   DNA polymerase II is involved in some types of DNA repair.
   DNA polymerase III is the principal replication enzyme in *E. coli*.

DNA POL I
NICK TRANSLATION ACTIVITY

DNA polymerase I is also endowed with **5'-3' exonuclease activity** involved in repair, recombination and replication activities, and in the elimination of the RNA primers that participate in DNA synthesis.



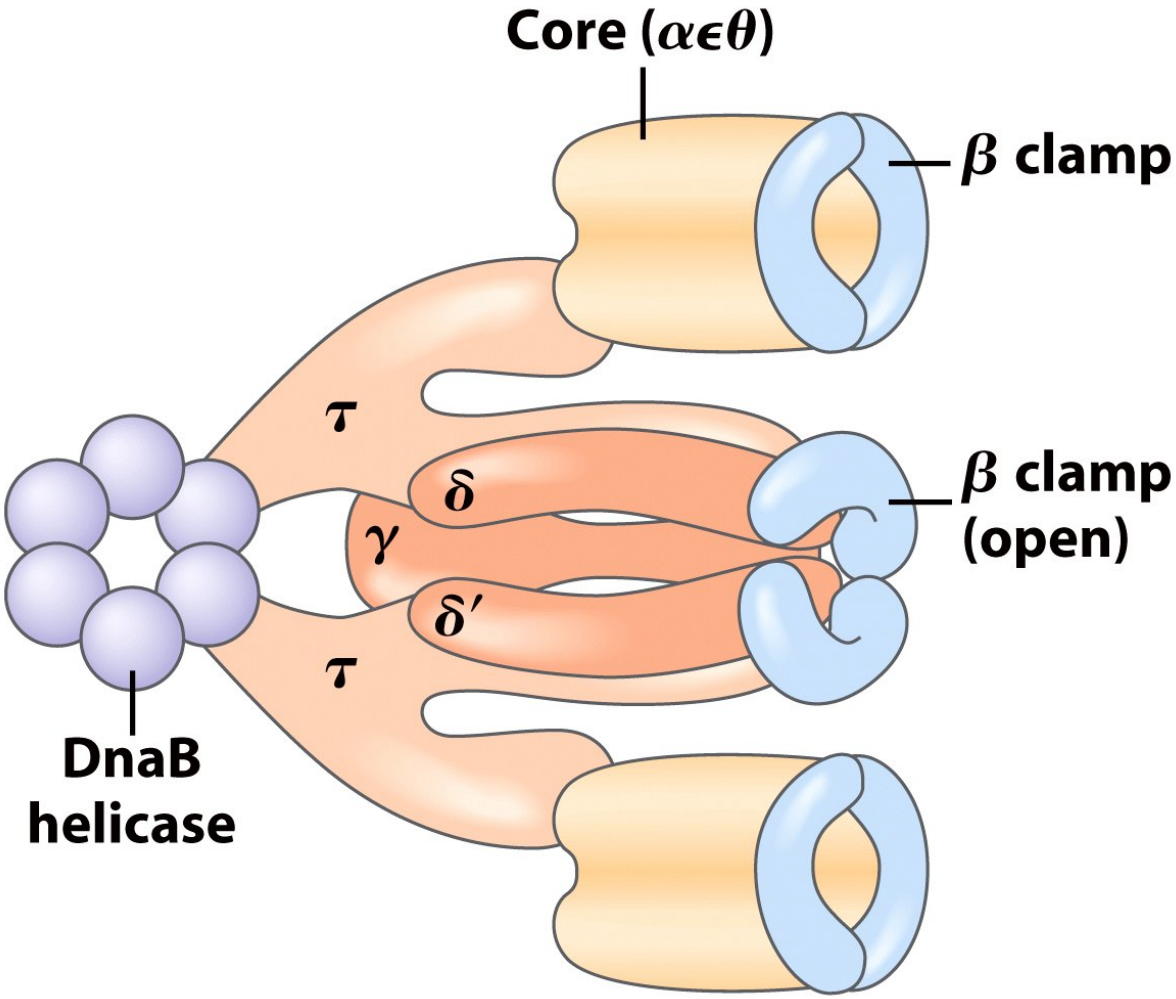
SUBUNITS OF DNAL POLYMERASE III OF *E. coli*

Subunit	Number of subunits per holoenzyme	M_r of subunit	Gene	Function of subunit	
α	2	129,900	*polC* (*dnaE*)	Polymerization activity	Core polymerase
ε	2	27,500	*dnaQ* (*mutD*)	3′→5′ Proofreading exonuclease	
θ	2	8,600	*holE*		
τ	2	71,100	*dnaX*	Stable template binding; core enzyme dimerization	Clamp-loading (γ) complex that loads β subunits on lagging strand at each Okazaki fragment
γ	1	47,500	*dnaX**	Clamp loader	
δ	1	38,700	*holA*	Clamp opener	
δ'	1	36,900	*holB*	Clamp loader	
χ	1	16,600	*holC*	Interaction with SSB	
ψ	1	15,200	*holD*	Interaction with γ and χ	
β	4	40,600	*dnaN*	DNA clamp required for optimal processivity	

*The γ subunit is encoded by a portion of the gene for the τ subunit, such that the amino-terminal 66% of the τ subunit has the same amino acid sequence as the γ subunit. The γ subunit is generated by a translational frameshifting mechanism (see Box 27–1) that leads to premature translational termination.

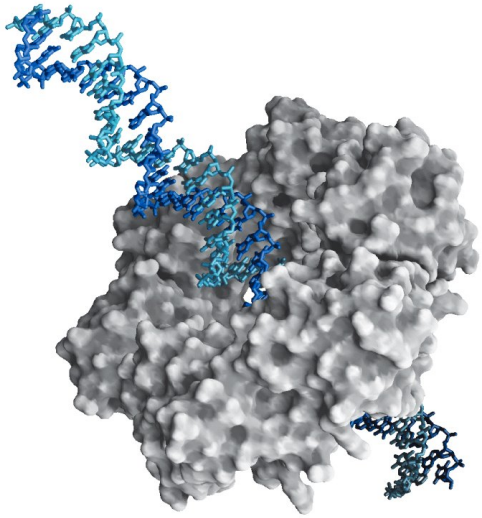
DNA polymerase III is much more complex than DNA polymerase I, having **ten types of subunits**. Its polymerization and proofreading activities reside in its α *and* ε *(epsilon) subunits, respectively.*

DNA polymerase III



The high processivity of DNA pol III is provided by the β ***subunits.***

The β *subunits associate in* pairs to form donut-shaped structures that encircle the DNA and act like clamps preventing the dissociation of DNA pol III from DNA.



The mechanism of DNA replication

1. Initiation

- Proteins bind to DNA and open up double helix.
- Prepare DNA for complementary base pairing.

2. Elongation

- Proteins connect the correct sequences of nucleotides into a continuous new strand of DNA.

3. Termination

- Proteins release the replication complex.

The events described below reflect information derived primarily from in vitro experiments using purified *E. coli* proteins, although the principles are highly conserved in all replication systems.

1. INITIATION OF DNA SYNTHESIS

PROTEINS REQUIRED TO INITIATE REPLICATION AT THE *E. coli* ORIGIN

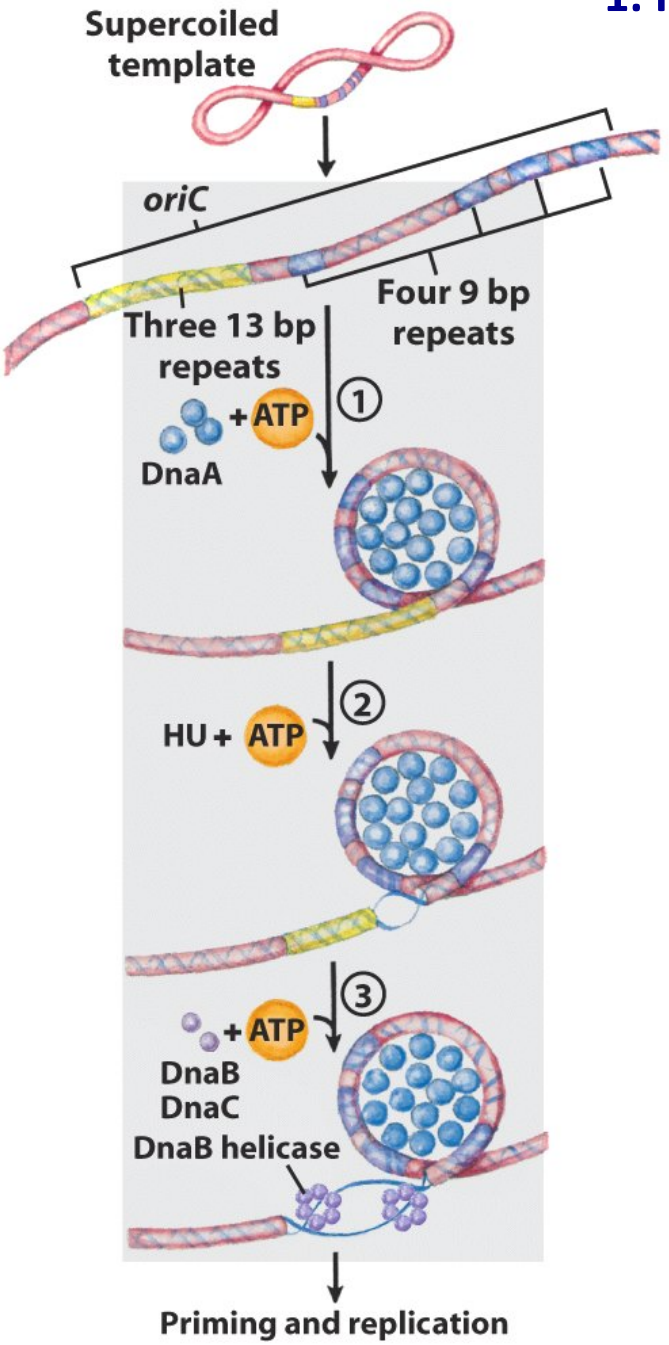
Protein	M_r	*Number of subunits*	*Function*
DnaA protein	52,000	1	Recognizes ori sequence; opens duplex at specific sites in origin
DnaB protein (helicase)	300,000	6*	Unwinds DNA
DnaC protein	29,000	1	Required for DnaB binding at origin
HU	19,000	2	Histonelike protein; DNA-binding protein; stimulates initiation
Primase (DnaG protein)	60,000	1	Synthesizes RNA primers
Single-stranded DNA–binding protein (SSB)	75,600	4*	Binds single-stranded DNA
RNA polymerase	454,000	5	Facilitates DnaA activity
DNA gyrase (DNA topoisomerase II)	400,000	4	Relieves torsional strain generated by DNA unwinding
Dam methylase	32,000	1	Methylates (5′)GATC sequences at *oriC*

*Subunits in these cases are identical.

1. At least nine different enzymes or proteins participate in the initiation phase of replication.

2. They open the DNA helix at the origin and establish a prepriming complex for subsequent reactions. The crucial component in the initiation process is the DnaA protein.

1. INITIATION OF DNA SYNTHESIS



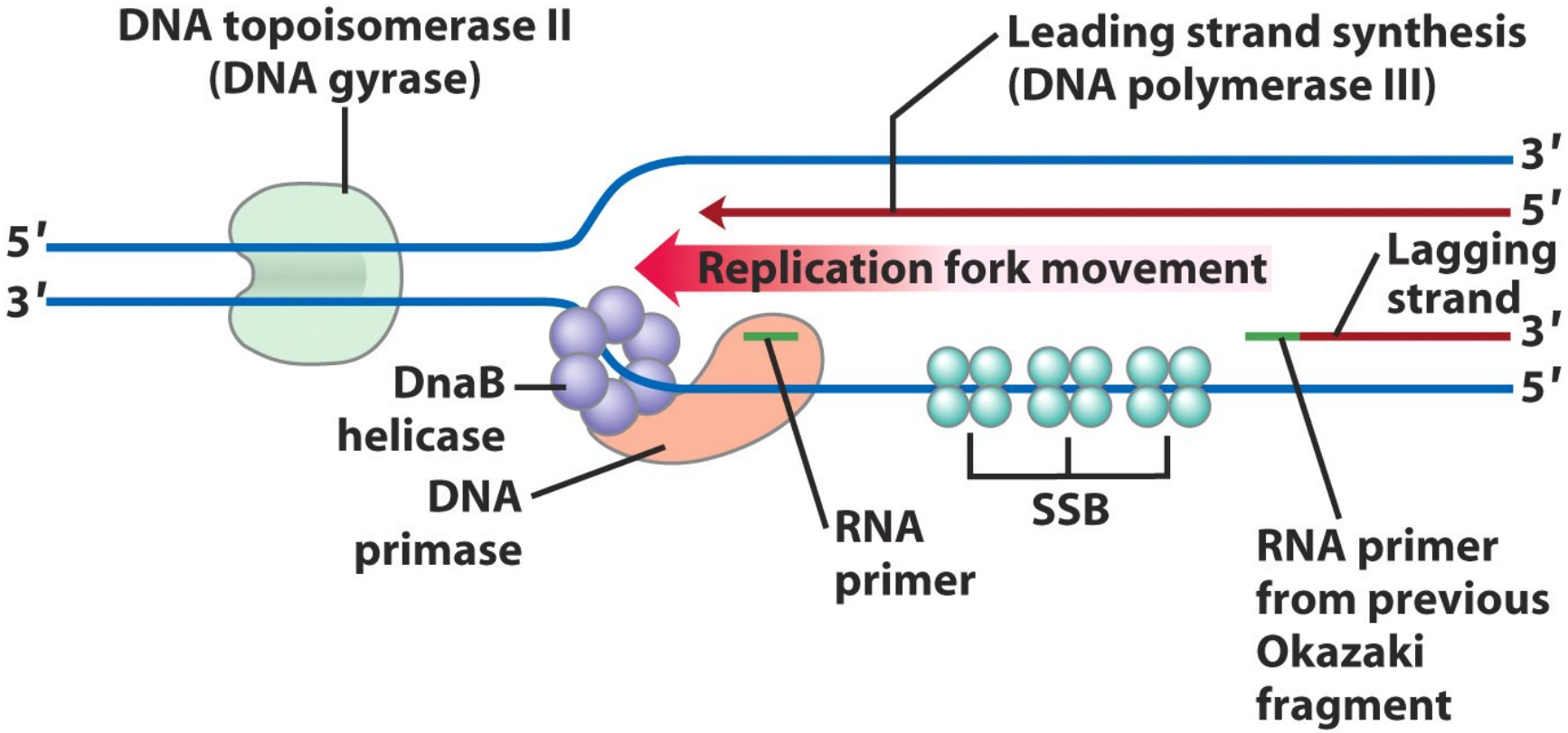
1. A single complex of about ten DnaA protein molecules binds to the four 9 bp repeats in the origin (step 1).

2. Then recognizes and successively denatures the DNA in the region of the three 13 bp repeats, which are rich in A=T pairs (step 2). This process requires ATP and the bacterial histone like protein HU.

3. Hexamers of the DnaB protein bind to each strand, with the aid of DnaC protein. The DnaB helicase activity further unwinds the DNA in preparation for priming and DNA synthesis.

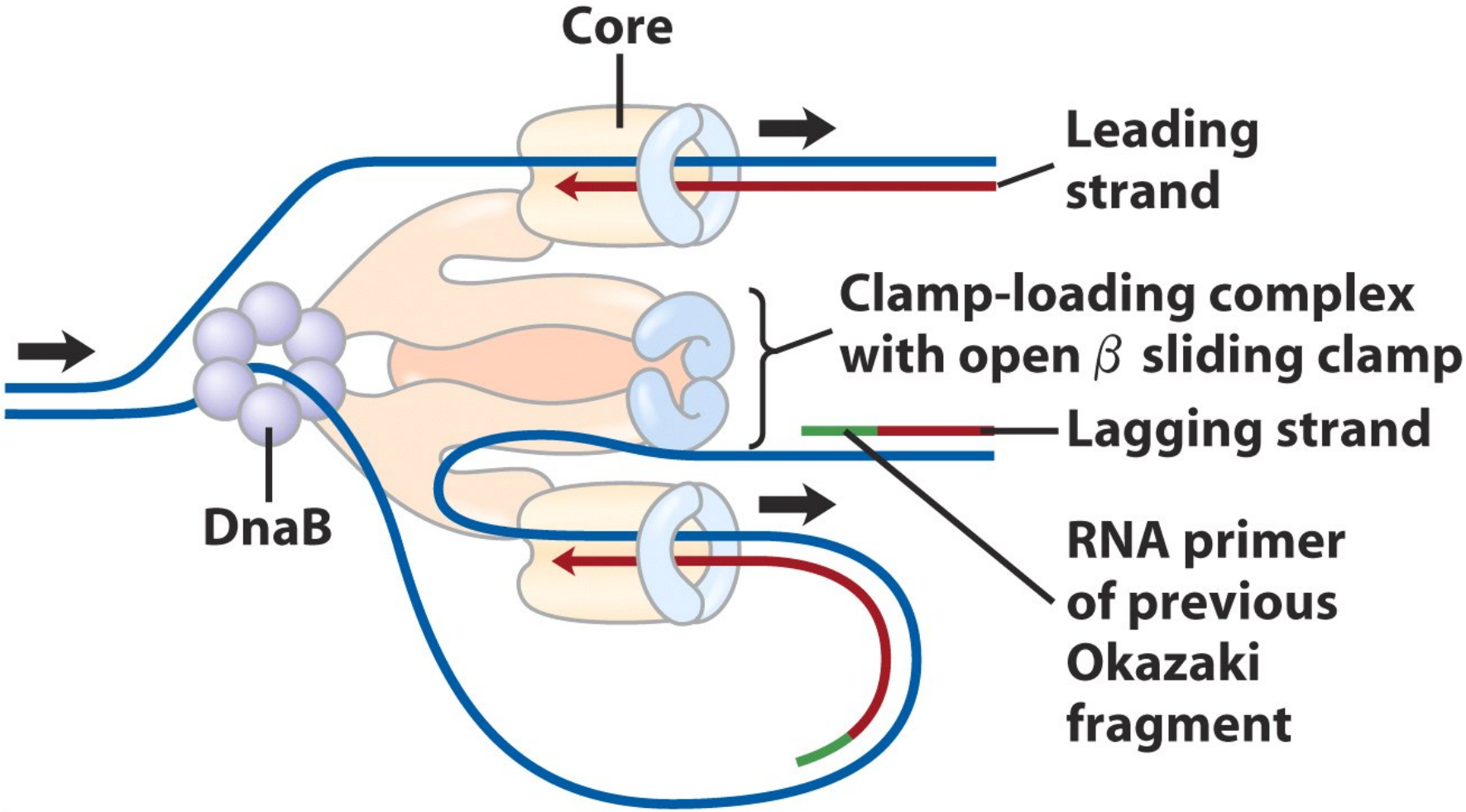
Binding of single stranded DNA binding protein (SSB) provides stability and prevents renaturation (not shown in the figure).

2. ELONGATION PHASE OF DNA SYNTHESIS



Leading strand synthesis begins with the synthesis by **primase** (DnaG protein) of a short (10 to 60 nucleotide) RNA primer at the replication origin. Deoxyribonucleotides are added to this primer by DNA polymerase III. Leading strand synthesis then proceeds continuously, keeping pace with the unwinding of DNA at the replication fork.

2. ELONGATION PHASE OF DNA SYNTHESIS

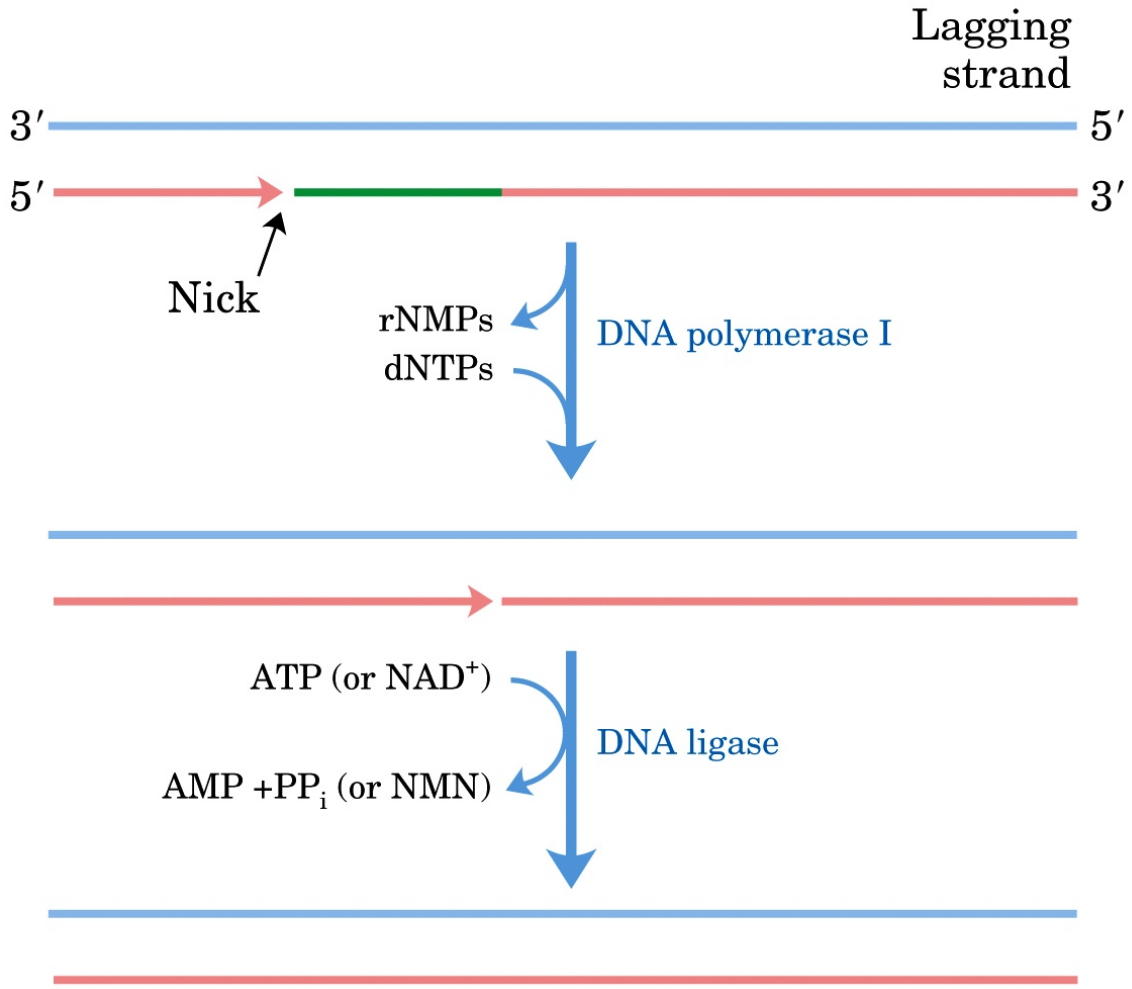


(a) Continuous synthesis on the leading strand proceeds as DNA is unwound by the DnaB helicase.

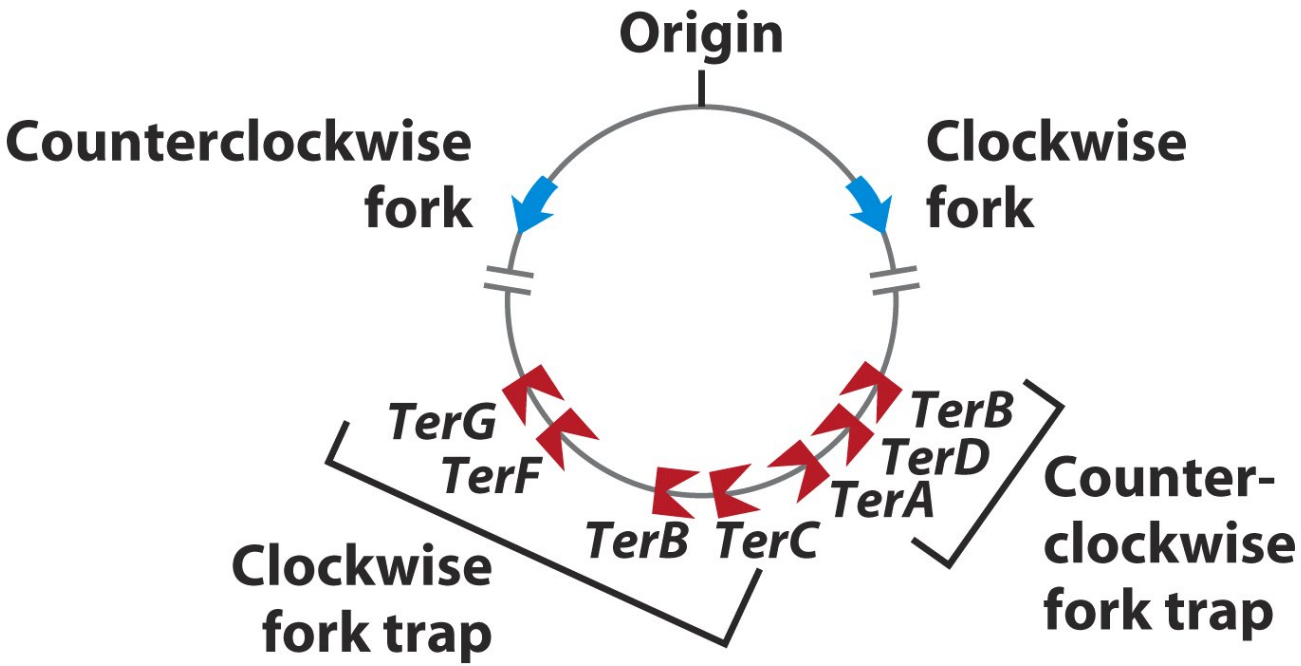
The DnaB helicase unwinds the DNA at the replication fork as it travels along the lagging strand template in the 5n3 direction.

2. ELONGATION PHASE OF DNA SYNTHESIS

DNA synthesis progresses at 1000 nucleotides per second in both strands. Once an Okazaki fragment has been completed, its RNA primer is removed and replaced with DNA by **DNA polymerase I**, and the remaining nick is sealed by **DNA ligase**.



3. DNA SYNTHESIS TERMINATION PHASE



Eventually, the two replication forks of the circular E. coli chromosome meet at a terminus region containing multiple copies of a 20 bp sequence called Ter (for *terminus).*

The Ter sequences are arranged on the chromosome to create a sort of trap that a replication fork can enter but cannot leave. The Ter sequences function as binding sites for a protein called Tus (*terminus utilization substance).*

DNA SYNTHESIS IN EUKARYOTES

- The **essential characteristics** of DNA replication are **conserved** between prokaryotes and eukaryotes. Many of the complexes are conserved both structurally and functionally.

- In **yeasts** there are several **origins of replication** per chromosome. These are sequences of aprox. 150 bp known as “autonomous replication sequences”: **ARS**.

- These origins of replication are bound by a multiprotein complex termed the **”origin of replication complex”** ORC. The ORC is bound and regulated by factors involved in the regulation of the **cell cycle**.

- DNA replication in human chromosomes occurs **bidirectionally** from **multiple origins af replication** separated by 30.000 to 300.000 base pairs.

DNA SYNTHESIS IN EUKARYOTES

Eukaryotic DNA polymerases

PROPERTIES OF EUKARYOTIC DNA POLYMERASES

	Polymerase α	Polymerase β	Polymerase δ	Polymerase ϵ	Polymerase γ	Polymerase ζ
Location	Nucleus	Nucleus	Nucleus	Nucleus	Mitochondria	Nucleus
3′–5′ Exonuclease Activity	No	No	Yes	Yes	Yes	No
Essential to Nuclear Replication?	Yes	No	Yes	Yes	No	No

Three eukaryotic DNA polymerases catalyze reactions in DNA replication, while others are involved in repair:

Pol α (**alpha**) synthesizes the RNA **primers** during **initiation**.

Pol δ (**delta**) performs the main task of concurrent **elongation** of both strands.
Its PCNA subunit, similar to β subunit in DNA pol III, increases processivity.

Pol ε (**epsilon**) is the other enzyme involved in nuclear DNA synthesis, possibly playing a role in eliminating the primers of Okazaki fragments in the lagging strand.

Pol γ (**gamma**) is encoded by a nuclear gene and synthetises **mitochondrial** DNA.